Government of Canada Gouvernement du Canada

V-03—Controlled conditions

Category: Volume

Bulletin: V-03 (rev. 4)

Document(s): W&M Regs, sections 2, 265-270

Issue date: 2013-04-01

Effective date: 2013-04-01

Supersedes: V-03 (rev. 3)

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1.0 Purpose

The purpose of this bulletin is to define under what conditions acceptance and in-service limits of error should be applied.

2.0 Scope

This bulletin applies to meters indicating in volumetric units of measure, when volumetrically tested in laboratories, dealers' or manufacturers' premises.

3.0 Background

Sections 265 to 270 of the Weights and Measures Regulations state that acceptance limits of error set out in column II of the appropriate tables apply only to tests made under controlled conditions. For tests made under other conditions, the in-service limits of error are set out in column III of those tables.

Pressure, temperature and flow rate are the primary factors which affect test results. In order to consider test conditions as "controlled" the effect of these factors must not contribute to an uncertainty in the test volume by an amount greater than ⅓ of the applicable limit of error for the device.

4.0 Policy

In order to evaluate the performance of a meter and apply acceptance limits of error as stated in column II of the tables in sections 265 to 270 of the Regulations, an inspector must be able to monitor the change in temperature, pressure and flow rate during testing. The limits are defined in part 4 below.

In the field, inspectors may not have means to adequately monitor or minimize the variations in temperature and pressure. There are situations where due to the nature of the changes to the device or the circumstances under which the device is used and tested, that it is not realistic to apply the acceptance LOE. The following provides examples of these situations and notes the LOE that should be applied:

- For trucks with new metering systems or recently repaired metering systems being initially inspected at the facilities where they were built or repaired, the in-service LOE should be used at all times. This is because they are complete systems being done under actual conditions of use, with the actual product they will be measuring and as such the testing should reflect the device's performance in normal use.
- When **new** dispensers are initially inspected in the factory or at dealers' premises, with substitute liquids, the acceptance LOE is to be applied. This is because the device will be installed at a different location, typically with a different pumping system and will measure different product than it was tested with. As such, it may not perform exactly as it did during the inspection. Appling the tighter acceptance LOE increases the chances that the device will perform within the in-service LOE once it is installed and has been broken in.
- In the case of self contained devices, such as dispensers, which require an initial inspection because the register has been changed, it is reasonable to apply the in-service LOE as this is the LOE that would be applied if the register had not been changed. In addition, there should be little change in the device's performance as the metering element will not change as a result of break-in.

The following defines the controlled conditions which apply to meters when tested to acceptance limits of error.

4.1 Temperature

4.1.1 The temperature stability, as measured in or near the meter, is within ±2 °C (of the mean) during any flow test.

4.1.2 The temperature differential of the test liquid, as measured in or near the meter and compared to that measured in the test measure, is within ±1 °C during any flow test.

4.1.3 When automatic temperature compensators (ATC) incorporated in meters are tested, the temperature, as measured in or near the meter, is within ±0.3 °C (of the mean) during any test of the compensator.

4.2 Pressure

4.2.1 The pressure of the product in the metering system is within ±100 kPa (of the mean) for normally liquid products and ±50 kPa for liquefied propane gas (LPG) during a single flow test.

4.3 Product composition

4.3.1 The composition of the product is not changed during the evaluation of a meter. The product used is the actual product the meter is intended to be used with or a test fluid selected in accordance with Bulletin V-09, "Inspection of Gasoline and Diesel Fuel Dispensers and Refuellers at Authorized Service Providers' or Dealers' Premises".

4.4 Environmental conditions

4.4.1 When using an open volumetric measure or prover, the dew point temperature of the air in the area where the tests are being performed is less than the temperature of the test liquid.

4.5 Flow tests

4.5.1 Flow Rate Variation: The volumetric flow rate is stable within a rate equal to ±10% of the maximum rated meter flow rate for 90% of the test volume.

4.5.2 Fast Flow Rate: Unless authorized by Measurement Canada's Engineering Services Directorate, the fast flow test is performed at a minimum of 80% of the maximum rated meter flow rate.

4.5.3 Slow Flow Rate: The slow flow test is performed at a flow rate greater than the minimum rated flow rate of the meter. To ensure the rate is above the minimum, the target flow rate should be determined by summing the following:

- **For all meters other than dispensers and refuellers**
  $(Minumum\ Approved\ Flow\ Rate) + (10\%\ of\ Minimum\ Approved\ Flow\ Rate)$
- **For dispensers and refuellers** $(Minimum\ Approved\ Flow\ Rate) + (5\ Litres/\min)$

4.6 Test volume

4.6.1 The test volume used is not less than:

a. 5 litres for slow flow meters with maximum flow rates of 10 L/min or less;
b. 20 litres for gasoline dispensers with maximum flow rates of 90 L/min;
c. 250 litres for refuellers with a maximum flow rate between 91 and 225 L/min (includes marine and truck refuellers and key locks).

4.6.2 For meters with a maximum flow rate in excess of 225 L/min, a volumetric measure with a standard nominal capacity greater than the volume displaced by the meter in a period of one minute should be used.

4.6.3 Shell Correction for Temperature: When a flow test requires that any size of volume standard is to be used to inspect devices against acceptance limits of error under controlled conditions, volume correction for the

expansion or contraction of the measure or prover shell is necessary in accordance to Bulletin V-17. The specialized test procedures provided in the Measurement Canada Field Inspection Manual may be referenced to calculate the shell correction.

4.6.4 Volume Correction for Pressure: When a meter is evaluated using a Vapour Displacement prover and the pressure differs by more than ±50 kPa from the vessel's base calibration pressure, the empirically derived correction factors, established at the time of most recent certification, are used to account for the change in test measures capacity.

4.6.4.1 When a meter is evaluated using a pipe prover and the pressure differs by more than, in the case of normally liquid products, ±100 kPa and, in the case of LPG, ±50 kPa from the vessel's base calibration pressure, a mathematical correction for the change in certified volume is applied. When corrections for the base volume of a pipe prover are necessary, the guidelines provided in the Measurement Canada training manual Operation and Use of Pipe Provers for Liquid Measurement will apply.

5.0 Revision

The original bulletin was issued on September 30, 1991, as Bulletin V-05.

5.1 The purpose of Revision 1 (Bulletin V-03) was to define the specific test conditions necessary for application of acceptance limits of error when testing in laboratories, dealers' or manufacturers' premises.

a. Revision 1 was re-sequenced to Bulletin V-03 and issued on November 30, 1995.

5.2 The purpose of Revision 2 was to clarify temperature corrections and measure shell corrections; revise the format and sequence; make editorial corrections with update to Measurement Canada and its directorates.

Revision 2 was issued September 1, 2007.

5.3 The purpose of Revision 3 was to simplify the requirements (PBRI2008) by revising sections to define controlled conditions applicable to meters when tested to acceptance limits of error; revise section 5.6.3 on “Shell Correction for Temperature” to meet policy stated in Bulletin V-17 (rev. 1) and section 5.3.1 on “Product Composition” to meet policy stated in Bulletin V-09 (rev. 3). Revision 3 was issued September 1, 2008.

5.4 The purpose of revision 4 is to provide examples of the applicable LOE for specific situations and add slow flow rate targets in section 4.5.3 to ensure meters are tested uniformly at flow rates slightly above their approved minimum flow rate. The terminology section was removed and subsections which followed were renumbered accordingly.

Date modified:
2017-03-28